Predicting bond dissociation energies using ALFABET – replacing QM calculations using the Schrödinger toolkit

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Introduction

Autoxidation is the degradation of a compound by molecular oxygen. The mechanisms of photochemically induced oxidations by singlet oxygen are topic of a separate work package and out of scope. The work package here is concerned with elucidation of the relative and absolute stabilities of atomic positions to autoxidation.
Autoxidation is a radical reaction pathway, and due to multiple reaction cycles before termination can lead to significant degradations. Starting points are typically trace impurities, often peroxides.

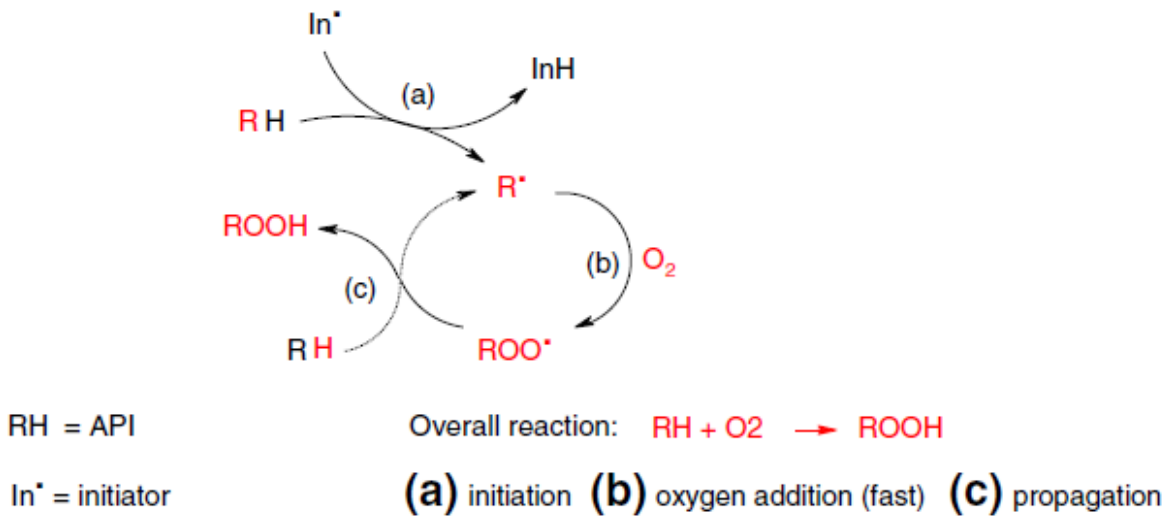


Figure 1: Illustration of the autoxidation process.

Breaking of a C-H bond is the first step in the cascade and calculation of bond dissociation energies (BDE) is reported to yield reliable predictions to characterize the reactivities of molecules.[1]

[1] Lienard, P., Gavartin, J., Boccardi, G. *et al.* Predicting Drug Substances Autoxidation. *Pharm. Res.,* **32**, 300–310 (2015). https://doi.org/10.1007/s11095-014-1463-7
Reid, D.L., Calvitt, C.J., Zell, M.T. *et al.* Early Prediction of Pharmaceutical Oxidation Pathways by Computational Chemistry and Forced Degradation. *Pharm. Res.,* **21**, 1708–1717 (2004). https://doi.org/10.1023/B:PHAM.0000041469.96466.12

Currently, the standard procedure for obtaining BDEs involves the Schrödinger toolkit, where quantum chemical (QM) calculations are performed. The homolytic BDEs defined by the enthalpy change for the following gas-phase reaction at 298 K:

Equation 1: Example of the homolytic bond dissociation reaction.

$$A - B \rightarrow A \cdot + B\cdot$$

However, obtaining such QM calculated BDEs is time-consuming and faster alternatives would be beneficial. In this report, we therefore evaluate a graph neural network (GNN) called ALFABET (**A** machine-**L**earning derived, **F**ast, **A**ccurate **B**ond dissociation **E**nthalpy **T**ool) from the Paton group for the prediction of homolytic BDEs.[2] The ALFABET model is accessible as a web application via https://bde.ml.nrel.gov/ and installation introductions for a local version of the ALFABET model are provided in the appendix.

The data used to develop the ALFABET model relies on QM computed BDEs, and an example of such a homolytic bond dissociation energy in the database with 63,229 molecules corresponding to 519,758 BDEs can be seen in Figure 2.

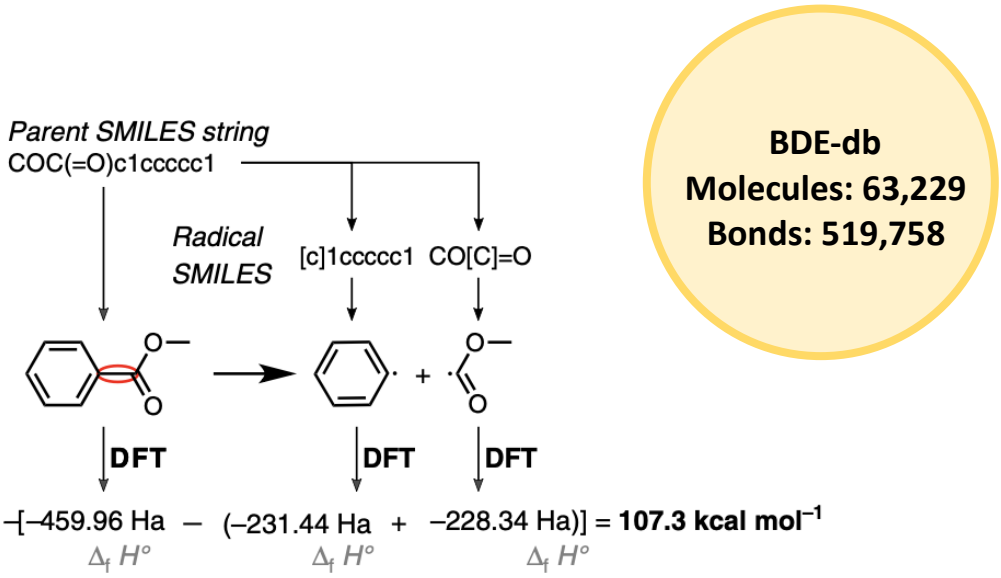


Figure 2: An example of an bond dissociation reaction in the BDE dataset used to train ALFABET.

While the Schrödinger toolkit performs calculations on the B3LYP/LACV3P** level of theory, the ALFABET database is based on the M06-2X/def2-TZVP level of theory given a benchmark

[2] St. John, P.C., Guan, Y., Kim, Y. *et al.* Prediction of organic homolytic bond dissociation enthalpies at near chemical accuracy with sub-second computational cost. *Nat. Commun.*, **11**, 2328 (2020). https://doi.org/10.1038/s41467-020-16201-z
St. John, P.C., Guan, Y., Kim, Y., Etz, B.D., Kim S., and Paton, R.S. Quantum chemical calculations for over 200,000 organic radical species and 40,000 associated closed-shell molecules. *Sci. Data.*, **7,** 244 (2020). https://doi.org/10.1038/s41597-020-00588-x
S. Sowndarya S. V., Y. Kim, S. Kim, P. C. St. John and R. S. Paton. Expansion of bond dissociation prediction with machine learning to medicinally and environmentally relevant chemical space. *Digit. Discov.*, **2**, 1900-1910 (2023). https://doi.org/10.1039/D3DD00169E

study of different DFT methods for a set of 368 experimentally measured BDEs from the *iBond* database (see Figure 3).[3]

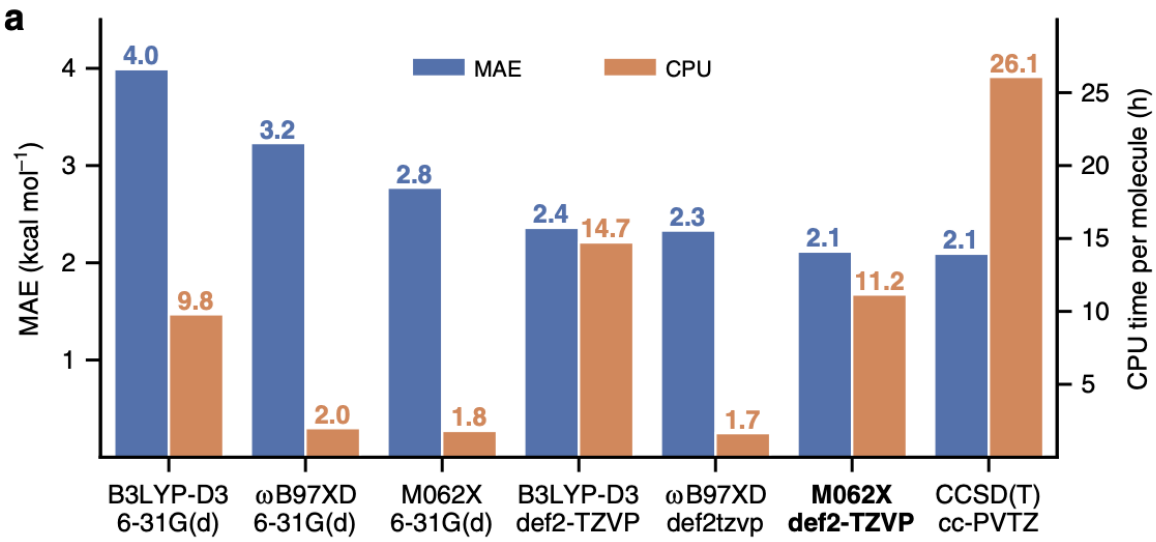


Figure 3: Trade-off between accuracy (left axis, blue) and computational cost (right axis, orange) for a selection of common QM methods. M06-2X/def2-TZVP was selected for subsequent calculations. MAE and CPU time were averaged over 368 different bonds with experimental measured BDEs from the iBond database.

The distribution of different bond types in the ALFABET database can be seen in Figure 4, and it highlights the chemical domain for which the ALFABET model can be employed.

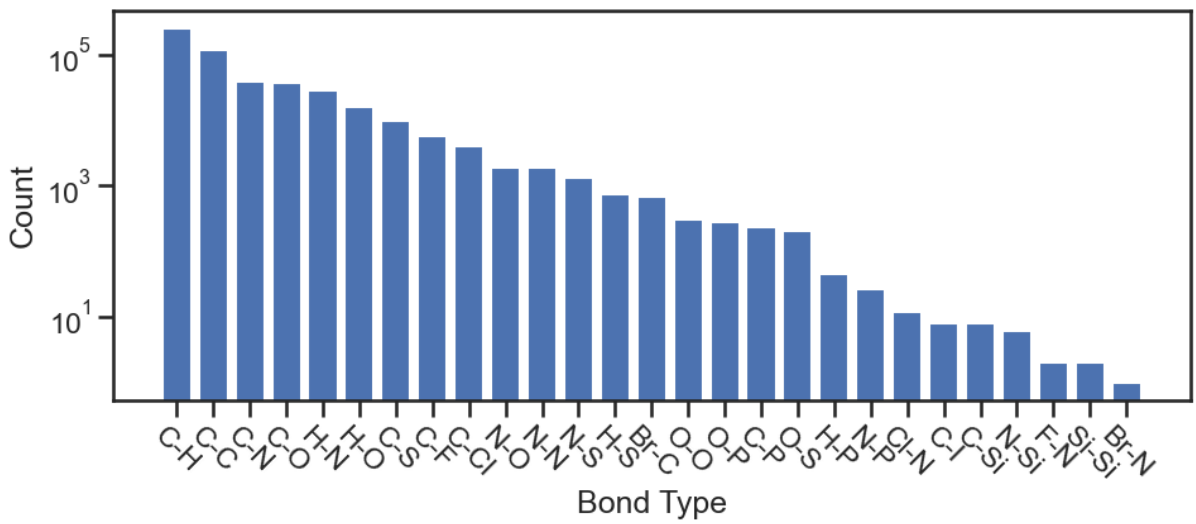


Figure 4: Distribution of different bond types in the ALFABET database.

[3] Internet Bond-energy Databank (pKa and BDE)—iBonD Home Page. http://ibond.nankai.edu.cn/

Results and Discussion

A machine learning model is only as good as the underlying data. In this first analysis, we therefore compare the results of the Schrödinger toolkit to the M06-2X/def2-TZVP level of theory used to construct the ALFABET database for the 964 molecules in the ALFABET test set (see Figure 5).

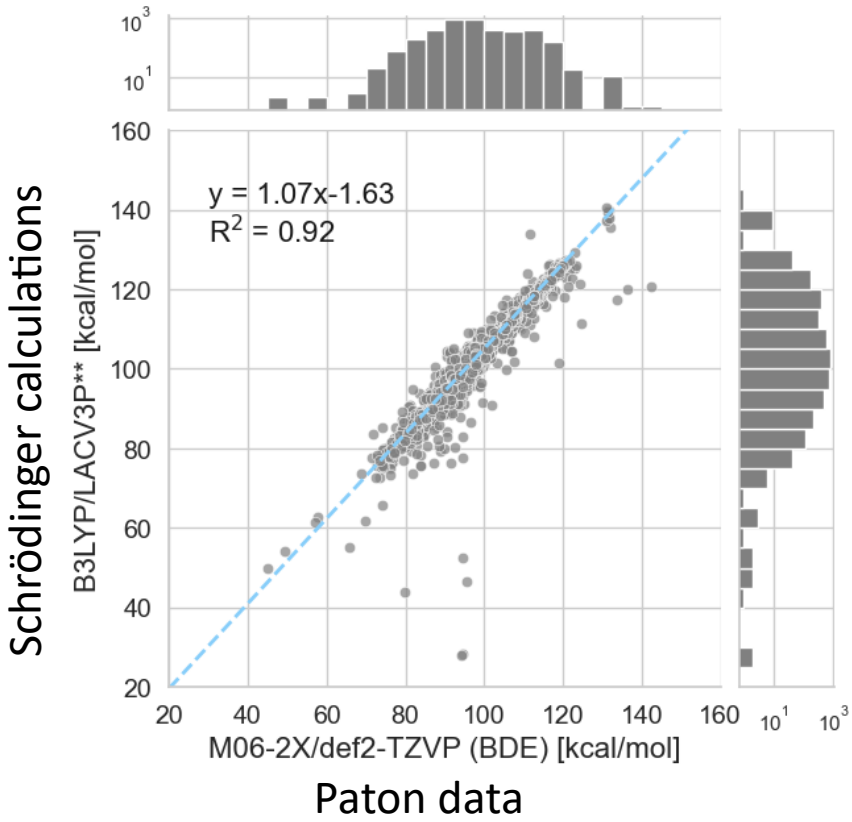


*Figure 5: Comparing C-H BDE results using the Schrödinger toolkit (B3LYP/LACV3P**) and the underlying DFT method (M06-2X/def2-TZVP) used to contruct the ALFABET dataset. The data consists of molecules from the ALFABET test set.*

Figure 5 shows a strong correlation between the two methods with an R^2 of 0.92. However, there are 12 outliers with an error >15 kcal/mol. When investigating the available structures of these outliers, we did not find any explanation for this discrepancy (see Figures 6 and 7). Note, that only 5 of the reactant structures are available at the M06-2X/def2-TZVP level of theory (Paton data). A complete comparison of all the structures is therefore not possible.

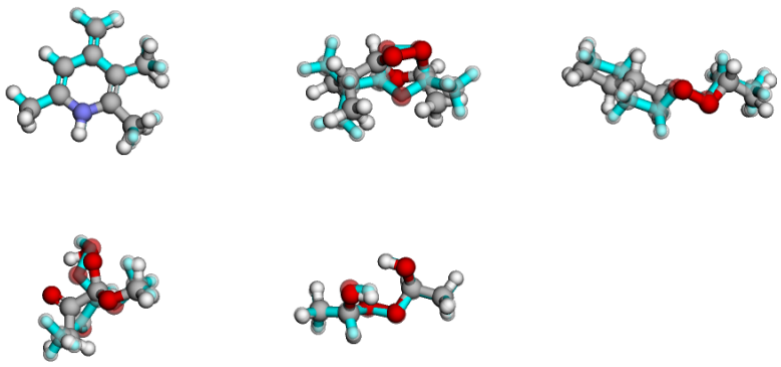


Figure 6: Outliers with an error >15 kcal/mol from Figure 5.

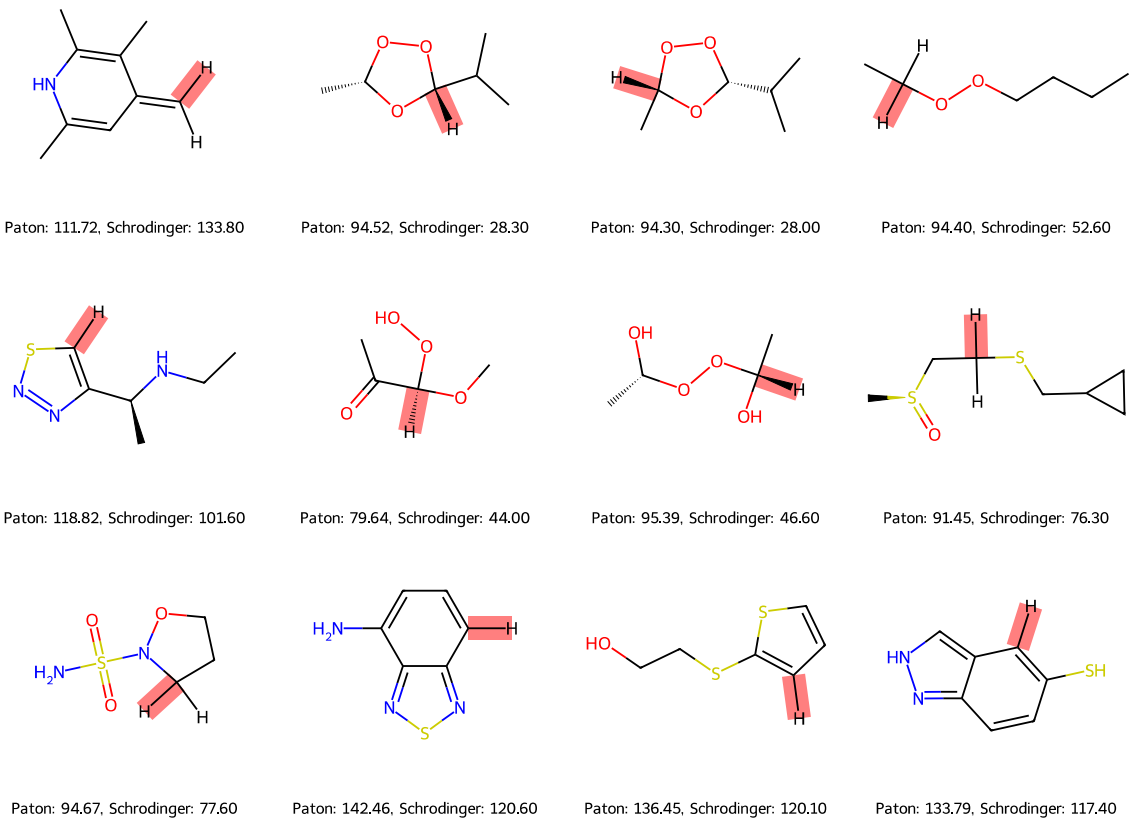


Figure 7: Outliers with an error >15 kcal/mol from Figure 5.

Nevertheless, the results in Figure 5 are quite satisfying, and we can now turn to the evaluation of the ALFABET model. The performance of the ALFABET model for C-H BDEs in the training, validation and test sets are shown in Table 1 and Figure 8.

Table 1: Performance of the ALFABET model for C-H BDEs in the training, validation, and test sets.

Dataset	Count (Bonds)	MAE [kcal/mol]
Train	247,482	0.32
Validation	4,076	0.55
Test	3,957	0.57

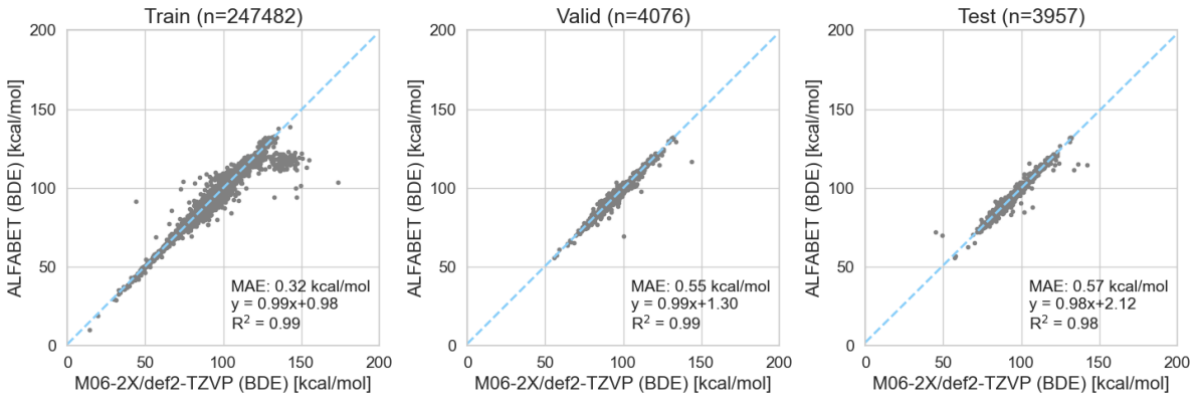


Figure 8: Performance of the ALFABET model for C-H BDEs in the training, validation, and test sets.

The results in Table 1 and Figure 8 show that the ALFABET model can achieve a mean absolute error (MAE) of <= 0.57 kcal/mol on all three data sets and a coefficient of determination (R^2) of >=0.98.

When testing the ALFABET model on 84 larger and more drug-like molecules, the ALFABET model yields a MAE of 0.61 kcal/mol compared to calculations at the M06-2X/def2-TZVP level of theory (see Figure 9). This shows that despite trained on smaller molecules, the GNN successfully predicts the BDEs for much larger species.

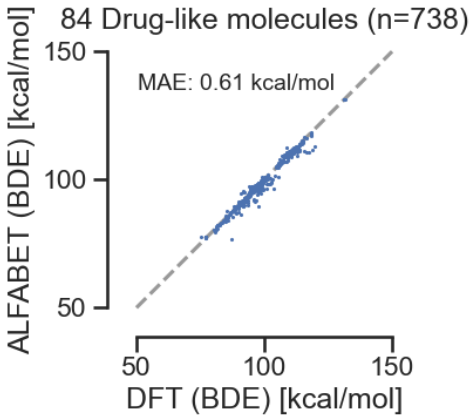


Figure 9: Comparing ALFABET model predictions to DFT calculations at the M06-2X/def2-TZVP level of theory for a test set of 84 drug-like molecules and a total of 738 C-H BDE values.

Moreover, the ALFABET model performs similarly to calculations at the M06-2X/def2-TZVP level of theory w.r.t. experimental BDEs from the iBond database as seen in Figure 10.

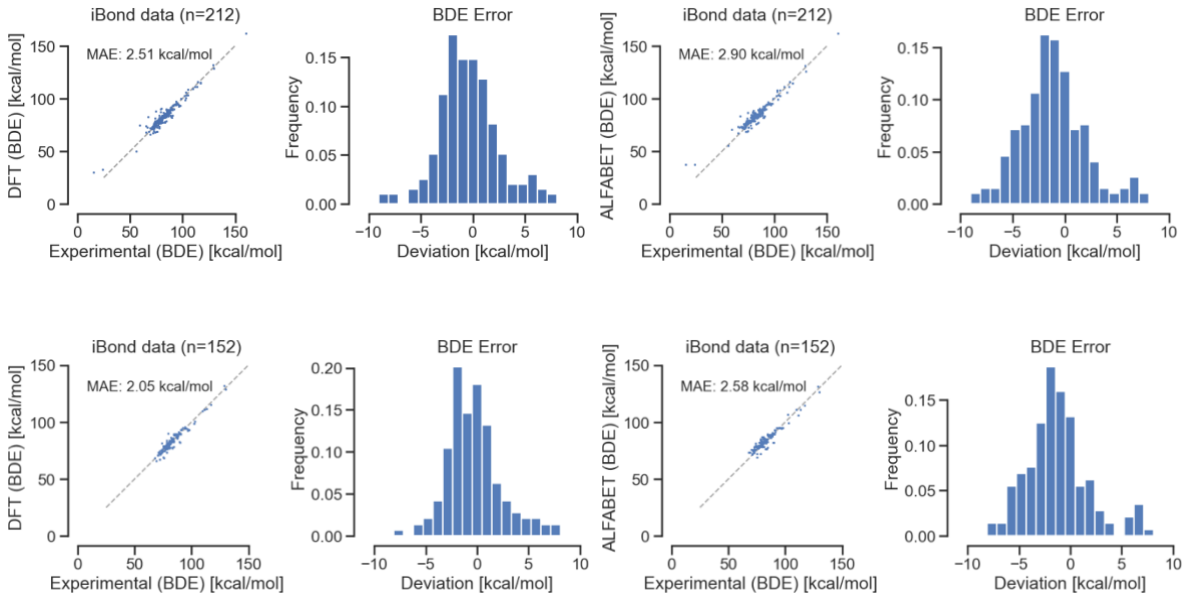


Figure 10: Comparing DFT calculations at the M06-2X/def2-TZVP level of theory and ALFABET model predictions to experimentally measured BDEs from the iBond database. The test set consist of 194 molecules with a total of 212 experimental C-H, C-C, O-H BDE values (152 C-H experimental BDE values) and bonds are not part of the training and validation set.

Testing ALFABET on out-of-sample ChEMBL data

Finally, we have constructed an out-of-sample data set with molecule from the ChEMBL database for which calculations using the Schrödinger toolkit have been performed. The dataset consists of 833 molecules, although 12 of these have been removed due to:

7 entries without Schrödinger BDE results.
2 duplicated entries with different ChEMBL ID.
4 negative Schrödinger BDE values, corresponding to 3 entries.

After removing these 12 molecules, the distribution of different bond types can be seen in Figure 11.

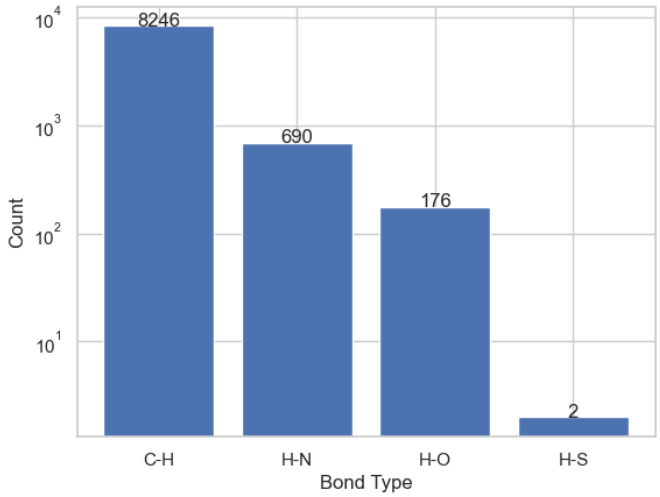


Figure 11: Distribution of different bond types in the out-of-sample ChEMBL dataset.

Now, Figure 12 shows the results of the Schrödinger toolkit and the ALFABET model. Again, the performance of the two approaches is quite similar with an R^2 of 0.92 when considering all the bond types shown in Figure 11 and an R^2 of 0.95 for only the C-H BDEs.

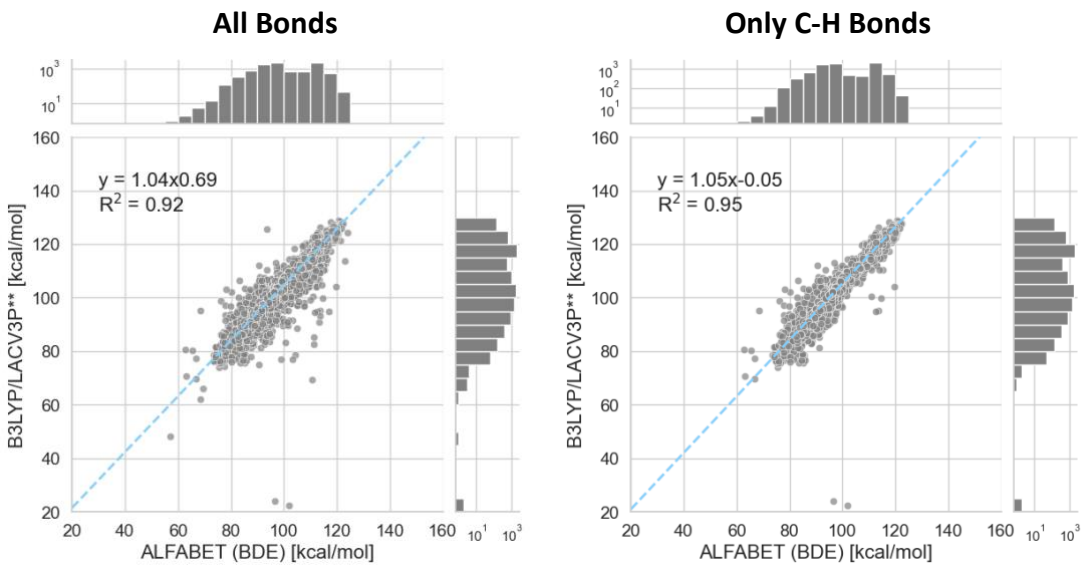


*Figure 12: Performance of the ALFABET model on the out-of-sample ChEMBL dataset compared to QM calculations using the Schrödinger toolkit (B3LYP/LACV3P**).*

Figure 13 shows the results of comparing the lowest BDE site(s) with ALFABET within a cutoff of 0-10 kcal/mol to the lowest BDE site with the Schrödinger toolkit. The reason for analysing these sites within a cutoff from 0-10 kcal/mol is that the different BDE sites can have similar BDEs. It is therefore difficult to get the same exact ordering of the BDEs. When simply looking at the lowest site for the two approaches (cutoff = 0 kcal/mol), the molecular accuracy is 73 %. However, with a cutoff of 3 kcal/mol on the ALFABET model predictions, the lowest BDE site with the Schrödinger toolkit is found 90 % of the times, which correponds to 737 out of the 821 molecules. When looking at how many sites are within the 3 kcal/mol, there are an average of 1.74 out of 11.10 BDE sites.

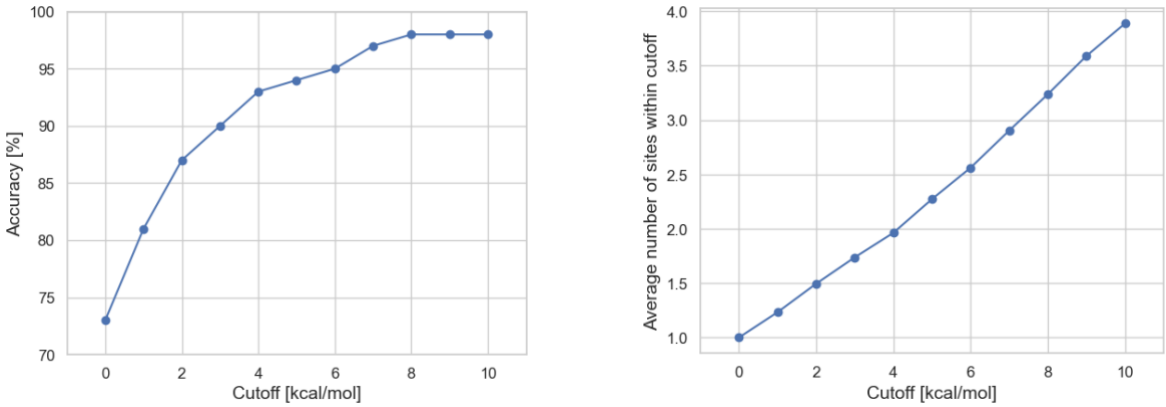


Figure 13: Comparing the lowest BDE site(s) with ALFABET within a cutoff to the lowest BDE site w.r.t. Schrödinger.

Conclusions and Outlook

A faster alternative to quantum chemistry (QM)-derived bond dissociation energies (BDEs) using the Schrödinger toolkit have been investigated. For this purpose, we have employed a GNN from the Paton group called ALFABET. The ALFABET model can perform on par with the QM calculations, while reducing the computational cost to a fraction of a second. We tested the ALFABET model against both experimental BDEs and QM-derived BDEs for both in- and out-of-sample test sets. These tests shows that the ALFABET can replace the computational expensive QM-calculations and could potentially be used to predict the site of autoxidation. Future work will collect a large set of molecules with experimental autoxidation data to test whether the predicted BDEs can be used to screen the chemical stability of a molecule with respect to autoxidation.

APPENDIX

Installation of ALFABET model:

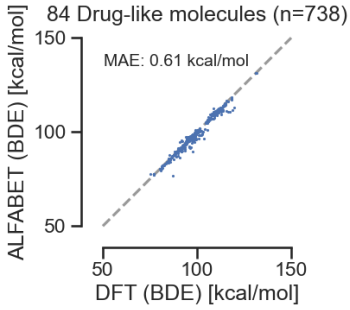
A folder called “setup_alfabet” should be downloaded via this link to the file management system at UCPH called ERDA: https://sid.erda.dk/sharelink/EkL35gvRxD

```
> cd setup_alfabet

> conda env create -f environment.yml

> conda activate alfabet

> sed -i 's\self._is_canon = False\self._is_canon = True\g'
~/anaconda3/envs/alfabet/lib/python3.8/site-packages/alfabet/fragment.py
# modify this, if anaconda3 is installed elsewhere

> cp -r pooch/ ~/.cache/. # On linux
> # cp -r pooch/ /Users/your_user_name/Library/Caches/. # On Mac (OBS!)

# test installation
> python
> import alfabet
> alfabet.__version__
'0.4.1'

> from alfabet import model
> df_pred = model.predict(['CC(=O)CN(C)C'])[['molecule', 'bond_index',
'bond_type', 'fragment1', 'fragment2',
'bde_pred']].sort_values(by='bde_pred')
> df_pred = df_pred[df_pred.bond_type.str.contains('H')] # only X-H
> print(df_pred)

# On my mac:
      molecule  bond_index bond_type fragment1                                          fragment2   bde_pred
5  CC(=O)CN(C)C           9       C-H       [H]  [H][C](C(=O)C([H])([H])[H])N(C([H])([H])[H])C(...  81.341064
6  CC(=O)CN(C)C          11       C-H       [H]  [H][C]([H])N(C([H])([H])[H])C([H])([H])C(=O)C(...  92.179237
4  CC(=O)CN(C)C           6       C-H       [H]  [H][C]([H])C(=O)C([H])([H])N(C([H])([H])[H])C(...  94.055733

# On new linux setup:
      molecule  bond_index bond_type fragment1                                          fragment2   bde_pred
5  CC(=O)CN(C)C           9       C-H       [H]  [H][C](C(=O)C([H])([H])[H])N(C([H])([H])[H])C(...  81.343048
6  CC(=O)CN(C)C          11       C-H       [H]  [H][C]([H])N(C([H])([H])[H])C([H])([H])C(=O)C(...  92.178696
4  CC(=O)CN(C)C           6       C-H       [H]  [H][C]([H])C(=O)C([H])([H])N(C([H])([H])[H])C(...  94.054970

> exit()
```

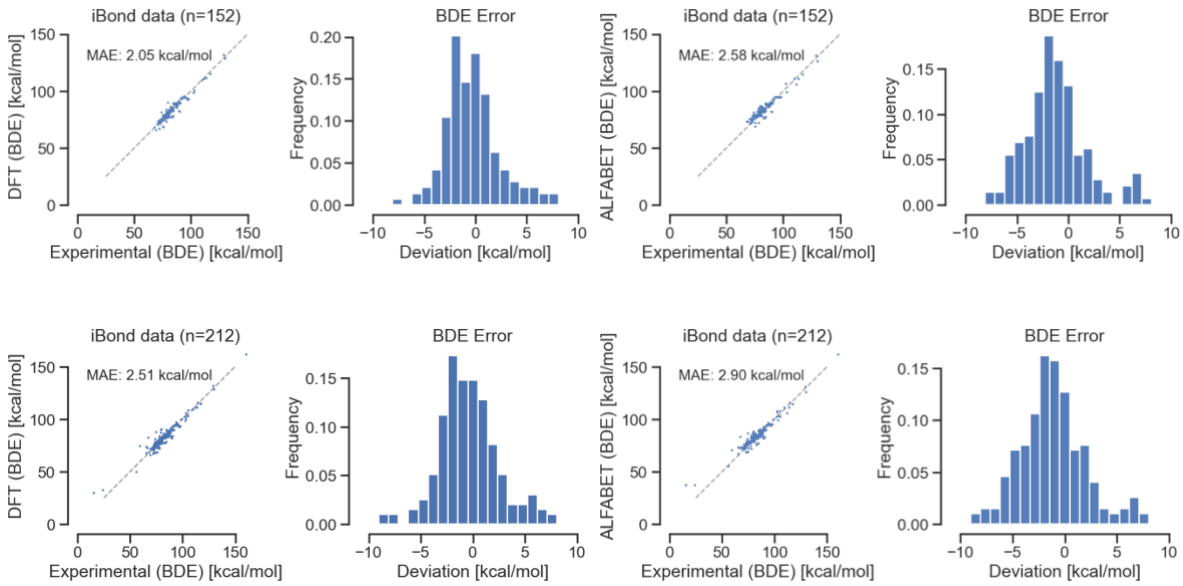
OBS! Always run predictions on canonical SMILES using:
df_pred = model.predict([Chem.MolToSmiles(Chem.MolFromSmiles(YOUR_SMILES))])

BDE test sets:

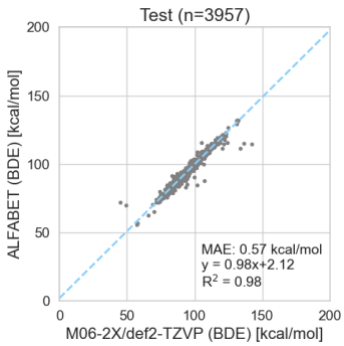
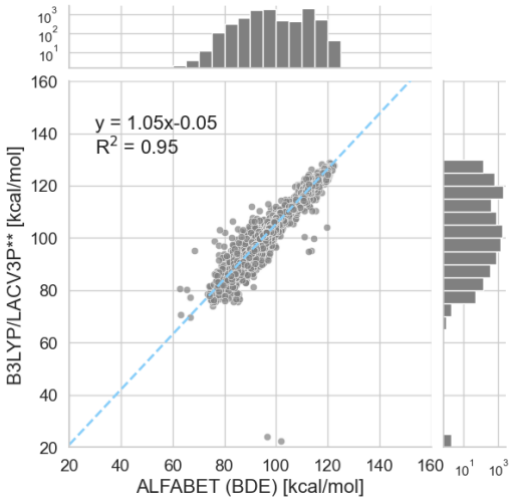
1) “druglike_cpds.csv”: A test set consisting of 84 drug-like molecules and a total of 738 C-H BDE values with calculated DFT (M06-2X/def2-TZVP) BDE values and predicted ALFABET BDE values.



2) “ibond_cpds.csv”: A test set consisting of 194 molecules from the iBonD database with a total of 212 experimental C-H, C-C, O-H BDE values (152 C-H experimental BDE values) including calculated DFT (M06-2X/def2-TZVP) BDE values and predicted ALFABET BDE values.



3) “paton_test_cpds.csv": A test set consisting of 964 molecules from https://github.com/pstjohn/alfabet-models/releases (alfabet-models/binary_files/bonds_for_neighbors.csv.gz) and a total of 3,957 C-H BDE values with calculated DFT (M06-2X/def2-TZVP) BDE values and predicted ALFABET BDE values.



4) ”chembl_diverse_subset.out.sdf”: An out-of-sample test set of 833 (821) ChEMBL molecules created by Andreas H. Göller to investigate how well the ALFABET model performs on out-of-sample molecules compared to QM calculations using the Schrödinger toolkit (B3LYP/LACV3P**).